adjustable arm grabs the steering wheel. Two extendable actuators attach to the brake and gas pedals, moving up and down to control them.

IVO monitors road conditions using a camera that removes the effect of background light so it is not blinded by the sun. The robot's cameras also look around to explore the environment. An additional camera faces the dashboard, so IVO can assess indicator lights and gauges for overheating, low gas and other problems.

IVO's simple design and straightforward functionality will make it easily affordable once production increases, according to Guterman. While it does not provide all the functionality of a vehicle built for autonomous driving, consumers can retrofit the $1600 unit into existing cars.

Guterman believes IVO might one day chauffeur people with disabilities, replace human drivers for safer and more efficient rescues under difficult conditions, and perhaps even replace truckers on long-haul routes.

Phones built for virtual reality

Google will launch at least eleven smartphones compatible with its own virtual reality (VR) platform called Daydream by the year-end. These Daydream-ready phones will be built for VR with high-resolution displays, ultra-smooth graphics and high-fidelity sensors.

Daydream View headset from Google

Google's Daydream is more like Samsung's Gear VR system rather than other VR platforms like the HTC Vive or Oculus Rift. This is because it does not require users to be tethered to a computer, but rather uses the user's smartphone to deliver the VR experience, together with the Daydream View headset. To experience VR content, users simply need to insert their smartphone into the Daydream View—a headset that is easy-to-wear, with a controller that is simple to use. The headset is made with soft, breathable fabric to help users stay comfortable. It is lightweight and fits comfortably over most eyeglasses. Plus, the facepad is removable to wash.

AI device wirelessly monitors sleep patterns at home

More than 50 million Americans suffer from sleep disorders. Diseases including Parkinson's and Alzheimer's can also disrupt sleep. Diagnosing and monitoring these conditions usually requires attaching electrodes and a variety of other sensors to patients, which can further disrupt their sleep.

The MIT device picks up RF signals reflected off sleeping subjects, which are then analysed to decipher sleep patterns (Image courtesy: Shichao Yue/MIT)

MIT researchers have devised a new way to monitor sleep stages without sensors attached to the body. A laptop-sized device bounces radio waves off the patient, and an artificial intelligence algorithm analyses the signals to accurately decode one's sleep patterns.

The device transmits RF waves at one-thousandth the power of Wi-Fi signals and picks up signals reflected from walls, furniture and sleeping subjects, whose tiniest movements change the frequency of the reflected signal. The deep neural network algorithm extracts the relevant sleep-related signals from the jumble of reflected signals and translates the data into meaningful sleep stages.

The system combines information on breathing, pulse and movements to decipher sleep stages—light, deep and rapid eye movement—with 80 per cent accuracy.

Computer software of the future

In a new Caltech study, researchers have demonstrated that quantum computing could be useful for speeding up the solutions to 'semidefinite programs'—a widely used class of optimisation problems. These programs include so-called linear programs, which are used, for example, when a company wants to minimise the risk of its investment portfolio or when an airline wants to efficiently assign crews to its flights.

The study presents a new quantum algorithm that could speed up solutions to semidefinite problems, sometimes